

A High Frame Rate Affordable Nystagmus Detection Method with Smartphones Used in Outpatient Clinic

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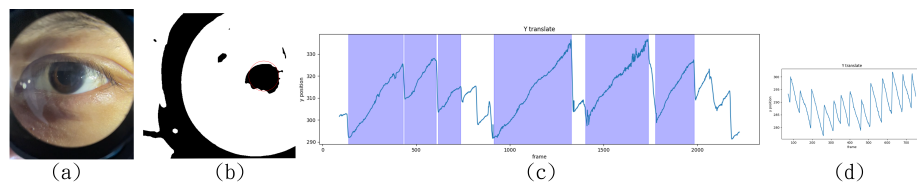


Figure 1: (a) Patient video recorded by smartphone through Frenzel goggles. (b) The blob after thresholding to capture the eyeball movement. (c)(d) The curves plotted by our programs, blue area in (c) represents the interval of nystagmus occurrence.

ABSTRACT

In this work, we present a high frame rate affordable nystagmus detection aid for outpatient clinic use. Using a Frenzel goggles and the high-speed video recording feature of a smartphone, high-frame rate eye videos of patients are obtained. Then, OpenCV morphological operation and blob detection library were modified to manually adjust the parameters according to the video to capture and track the eye center movements. Next, we correct the error caused by hand shaking during photography and predict the eyeball movement during blinking. Finally, with rule-based detecting method, nystagmus was detected and classified based on the modified tracking data. In our experimental, we collaborated with a professional ENT doctor to obtain multiple high-frame rate eye videos of patients. Experimental results show that this procedure can provide preliminary diagnosis results for physicians to refer to, but the final diagnosis needs to be made by a professional ophthalmologist. Plus, this method has been used by the ENT doctor to record the result simultaneously.

CCS CONCEPTS

• **Applied computing** → *Health care information systems.*

KEYWORDS

nystagmus detection, eye movement, AR, client server model

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1 INTRODUCTION

Nystagmus is an involuntary rhythmic movement of the eye, which is often used as a diagnostic basis for vertigo, dizziness and other related diseases. About 0.3% population has this problem annually, for people over 60 years old, it can reach 30%; furthermore, it has an annual recurrence rate of 30%, causing a lot of trouble to patients. Jerk nystagmus is a typical type of nystagmus which consists a slow phase and a quick phase. Traditionally, Videonystagmography (VNG) is used to diagnose nystagmus. However, due to the high cost of this device, only large regional hospitals can afford it, and patients have to wait for several days to do VNG examination, so it cannot be used in outpatient clinic to achieve the goal of early diagnosis. In our earlier study, the mixed-reality wearable device HoloLens2TM is proposed to record eyeball movement as nystagmus detection. Although the price of HoloLens2TM is much cheaper than VNG, the sampling rate of HoloLens2TM can only be up to 60 frames per second, so it is unable to collect higher frame rate eyeball movement data.

We aim to provide a convenient, affordable and high sampling rate nystagmus diagnostic procedure to improve the above shortcomings. In terms of convenience and affordability, we choose to use smartphones and Frenzel goggles which are commonly used by medical doctors for recording eyeball movement. As for high sampling rate, we use the high-speed recording feature of the iPhone Pro 12 to record video. Its sampling rate can be up to 240 frames

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per second, therefore can provide high frame rate eyeball movement data. For detection, we use OpenCV, a public computer vision library, to capture the eye center movement with suitable modification. Finally, the tracking data after post-processing are tested to determine whether the subject has nystagmus symptoms, but it still needs a professional ENT doctor (otolaryngologist) to verify and make the final diagnosis based on the results. We hope that the public can obtain preliminary diagnosis results and seek medical treatment as early as possible through this convenient diagnostic procedure, since it is also a client-server model.

2 PROPOSED METHOD

In our proposed workflow, it is divided into four parts: high-frame rate recording, detection of eyeball movement, error compensation and post-processing, and diagnosis of nystagmus. First, we took advantage of the 240 FPS high-speed recording feature of the iPhone 12 Pro to record the eyeball movement video with high frame rate.

2.1 Detection of Eyeball Movement

We use the OpenCV library to get the movement of the eyeball in the video. First, for each frame, we do binarization and opening operation to obtain binary image. Then apply a median blur to the binary image to get a smoother result. Since the ambient light and shooting quality of each video are not necessarily the same, the threshold of binarization needs to be manually adjusted to filter out the eyeball. Then we use OpenCV blob detection module to detect the location of black area. We set the circularity, inertia ratio, convexity and area settings of the module to filter out the eyeball. As there may be more than one blob detected, we choose the blob with the largest area. Then, with the blob center as the eyeball center, the X and Y pixel coordinates of the blob center are recorded to obtain the position information of the eyeball movement.

2.2 Error Compensation and Post-processing

In post-processing, we remove the outliers caused by the misdetection of blobs through calculating the distance from the median of coordinates data. Besides, there is a small amount of missing data due to the blinking of subject or the change in brightness of the video, so the eyeball or goggles are not successfully detected in every frame. We use prediction methods such as Kalman Filter and linear interpolation to estimate the coordinate at the frame where the detection fails, thus ensuring that every frame has coordinate data. As the shaking of the hand during recording will affect the result, we need to obtain the displacement caused by the shaking of the smartphone to correct the error. As the Frenzel goggles will not move during recording, we can get the movement of the smartphone through the displacement of the Frenzel goggles. The ideas of detecting Frenzel goggles is basically the same as detecting an eyeball. Finally, the relative displacement caused by the hand shaking is compensated to obtain the correct eyeball movements.

2.3 Diagnosis of Nystagmus

With the trajectory data, we can plot the displacement curve using frame number (time index) as the X-axis and the eyeball trajectory coordinate on the Y-axis, and we set a moving window to scan from the first frame to the last frame, to see if the curve contains a jerk

nystagmus curve. We take the coordinates and frame numbers of the head p_h, f_h and tail p_t, f_t of the window and calculate the slope S of the line connecting the two points by the following formula:

$$S = \frac{4.8(p_t - p_h) \cdot \alpha \cdot d}{f_t - f_h} \quad (1)$$

where α is the frame rate of the video, d represents the length of a pixel in the real world. When detecting the Frenzel goggles, the diameter of the lens can be obtained. Since the real diameter of the Frenzel goggles is known, we can calculate the ratio between the pixel and the real length to get d . The unit of S is degrees/second. Then we set a threshold, and if S was greater than or equal to this threshold, implies the occurrence of nystagmus in this window. Finally, we combine the overlapping regions of these windows to obtain the interval of nystagmus and mark them on the curve plot.

3 EXPERIMENTAL RESULTS

In practice, we develop with Python 3.9 and our programs run on Intel® Core™ i9-10900K 5.30GHz. Our average processing speed is 30 frames per second for a 240FPS 960x540 resolution video. Therefore, a typical video clip from a patient will take only a few seconds to get the final prediction results. The diameter of Frenzel goggles are 5.2 and 4.8 cm. The window length and slope threshold are adaptive values such as 100 frames and 9 degrees/sec. The results show that we can successfully detect and label nystagmus in our patients, as shown in Figure 1(c). This curve shows a typical jerk nystagmus, and the program successfully identifies and marks the slow phase of them. The precision of our method is 95%, and the accuracy and the recall are 93% and 95%, respectively, based on 57 videos recorded from 34 patients. As mentioned before, the recording quality of the video will greatly affect the detection results, and the failure cases actually will happen less than 3% of our tests, and will not cause troubles to the medical doctors.

4 CONCLUSION AND FUTURE WORKS

We propose a new semi-automatic nystagmus diagnostic procedure, hoping to improve the shortcomings of traditional methods. In the experiment, nystagmus can be successfully diagnosed in most cases, and the reliability of the results can be confirmed by professional ENT doctors. Our co-authors are MDs from national medical centers, and therefore, comparison can be made with traditional methods. However, the quality of the recorded video has certain requirements, and the accuracy of the diagnosis may be affected by some external factors. The future improvement will focus on the automatic selection of binary threshold to achieve the goal of automatic detection and establish a client server model on the internet, providing an anytime, anywhere and affordable diagnostic service. There is a possibility that even the Frenzel goggles is not necessary in this process. We believe that if this program can be successfully applied to primary care, it may even revolutionize the way nystagmus is diagnosed in the future.

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